

```
!pip install transformers
!pip install datasets
!pip install torch
!pip install scikit-learn
```

```
Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: tokenizers<0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: typer-slim in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: hf-xet<2.0.0,>=1.2.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: typing-extensions>=4.1.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: datasets in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: pyarrow>=15.0.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: dill<0.3.9,>=0.3.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: requests>=2.32.2 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: tqdm>=4.66.3 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: xxhash in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: multiprocessing<0.70.17 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: fsspec<=2025.3.0,>=2023.1.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: huggingface-hub>=0.24.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: aiohttp!=4.0.0a0,!4.0.0a1 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: hf-xet<2.0.0,>=1.2.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: typing-extensions>=4.1.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages
```

```
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages/pytz-2022.7 (from transformers==4.34.0)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages/tzdata-2022.7 (from transformers==4.34.0)
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in /usr/local/lib/python3.12/dist-packages/aiohappyeyeballs-2.5.0 (from aiohttp==3.9.1)
Requirement already satisfied: aiosignal>=1.4.0 in /usr/local/lib/python3.12/dist-packages/aiosignal-1.4.0 (from aiohttp==3.9.1)
Requirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.12/dist-packages/attrs-24.2.0 (from aiohttp==3.9.1)
```

```
import torch
import pandas as pd
import numpy as np

from datasets import load_dataset
from transformers import DistilBertTokenizerFast
from transformers import DistilBertForSequenceClassification
from transformers import Trainer, TrainingArguments

from sklearn.metrics import accuracy_score, precision_recall_fscore_support
```

```
dataset = load_dataset("imdb")
```

```
train_dataset = dataset['train']
test_dataset = dataset['test']
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning: The secret `HF_TOKEN` does not exist in your Colab secrets.
```

```
To authenticate with the Hugging Face Hub, create a token in your settings tab. You will be able to reuse this secret in all of your notebooks.
```

```
Please note that authentication is recommended but still optional to access public datasets. warnings.warn(
```

```
README.md: 7.81k/? [00:00<00:00, 502kB/s]
```

```
Warning: You are sending unauthenticated requests to the HF Hub. Please set a token as an environment variable 'HF_TOKEN' or as a secret in your Colab settings. WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated
```

```
plain_text/train-00000-of-00001.parquet: 100% 21.0M/21.0M [00:00<00:00, 105MB/s]
```

```
00001.parquet: 100%
```

```
plain_text/test-00000-of-00001.parquet: 100% 20.5M/20.5M [00:00<00:00, 102MB/s]
```

```
00001.parquet: 100%
```

```
plain_text/unsupervised-00000-of-00001.p(...): 100% 42.0M/42.0M [00:00<00:00, 109MB/s]
```

```
00000-of-00001.p(...): 100%
```

```
Generating train split: 100% 25000/25000 [00:00<00:00, 91019.01 examples/s]
```

```
Generating test split: 100% 25000/25000 [00:00<00:00, 97549.05 examples/s]
```

```
Generating unsupervised split: 100% 50000/50000 [00:00<00:00, 129633.37 examples/s]
```

```
print(train_dataset[0])
```

```
{'text': 'I rented I AM CURIOUS-YELLOW from my video store because of all the
```

```
{
  'text': "This movie was fantastic...",
  'label': 1
}
```

```
{'text': 'This movie was fantastic...', 'label': 1}
```

```
tokenizer = DistilBertTokenizerFast.from_pretrained('distilbert-base-uncased
```

```
tokenizer_config.json: 100% 48.0/48.0 [00:00<00:00, 4.43kB/s]
```

```
vocab.txt: 232k/? [00:00<00:00, 7.70MB/s]
```

```
tokenizer.json: 466k/? [00:00<00:00, 6.70MB/s]
```

```
def tokenize_function(example):
    return tokenizer(example['text'], padding="max_length", truncation=True)
```

```
train_dataset = train_dataset.map(tokenize_function, batched=True)
```

```
test_dataset = test_dataset.map(tokenize_function, batched=True)
```

```
Map: 100% 25000/25000 [00:28<00:00, 981.44 examples/s]
```

```
Map: 100% 25000/25000 [00:26<00:00, 1011.82 examples/s]
```

```
print("Training samples : 25,000")
```

```
print("Testing samples : 25,000")
```

```
Training samples : 25,000
```

```
Testing samples : 25,000
```

```
train_dataset.set_format(type='torch', columns=['input_ids', 'attention_mask'
```

```
test_dataset.set_format(type='torch', columns=['input_ids', 'attention_mask'
```

```
model = DistilBertForSequenceClassification.from_pretrained(
    "distilbert-base-uncased",
    num_labels=2
)
```

```

config.json: 100% 483/483 [00:00<00:00, 38.9kB/s]
model.safetensors: 100% 268M/268M [00:04<00:00, 153MB/s]
Loading weights: 100% 100/100 [00:00<00:00, 417.84it/s, Materializing param=distilbert.transforr
DistilBertForSequenceClassification LOAD REPORT from: distilbert-base-uncased
Key | Status |
-----+-----+

```

Start coding or [generate](#) with AI.

```
vocab.transform.weight | UNEXPECTED |
```

```

def compute_metrics(pred):
    labels = pred.label_ids
    preds = pred.predictions.argmax(-1)

    precision, recall, f1, _ = precision_recall_fscore_support(labels, preds,
    acc = accuracy_score(labels, preds)

    return {
        'accuracy': acc,
        'f1': f1,
        'precision': precision,
        'recall': recall
    }

```